Vane pump

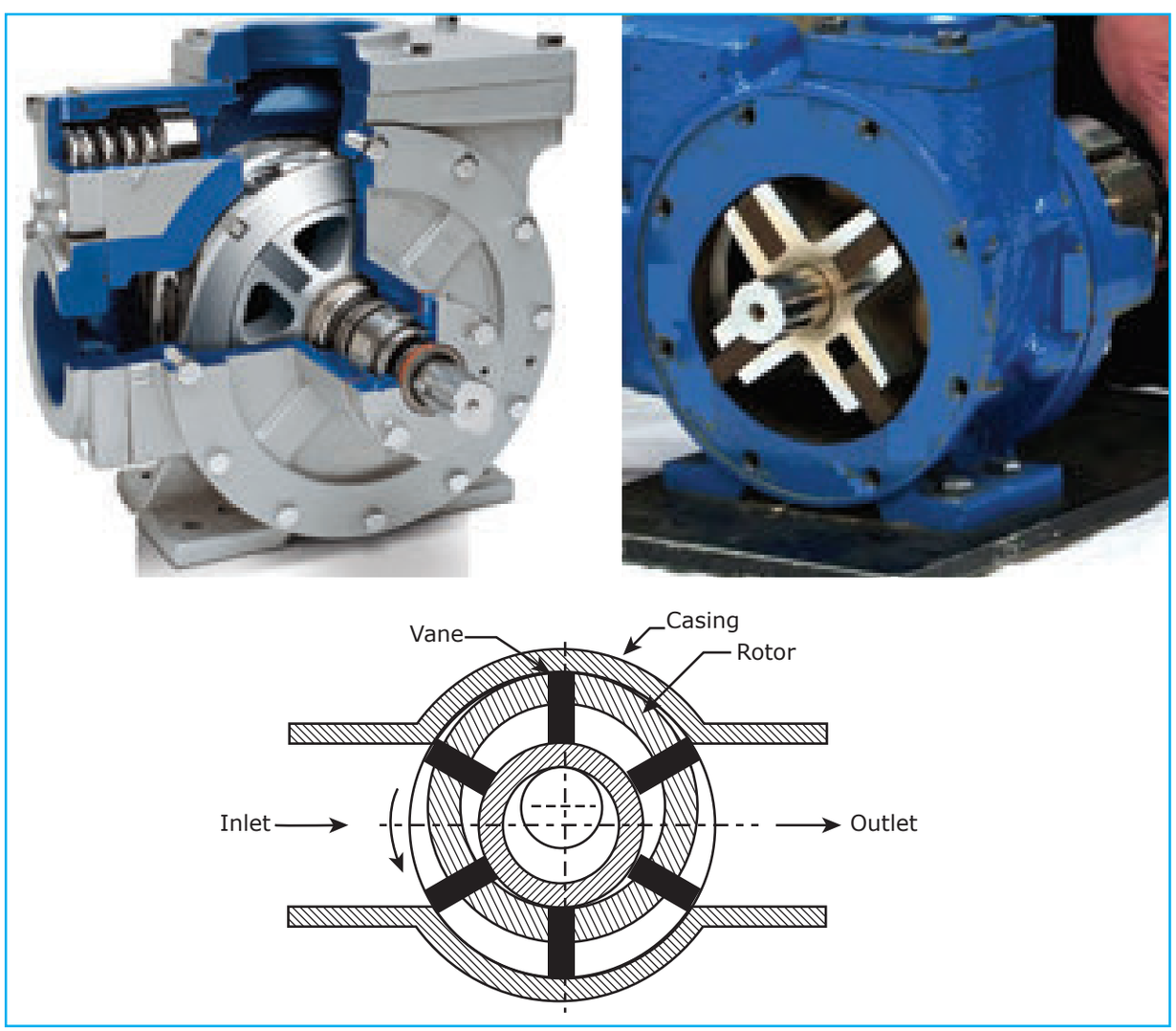
Construction

Vane pumps are equipped with a rotor. Radical slots are present on the rotor. Vanes are attached to these slots and are designed to slide inside them. Rotor is place inside the casing. The casing and the rotor are placed eccentric. When the rotor is made to rotate, vanes come out of the slots due to centrifugal force. Vanes rotate by touching the inner walls of the casing. In some pumps, vanes are attached to the slots by means of springs to increase the pressure. A vane pump is illustrated in figure.

Working principle of vane pump

The rotor is rotated by a prime mover. The liquid is trapped in between the vanes. The design of the pump is such that the volume of the incoming liquid is increasing and the volume of the outgoing liquid is decreasing. So the liquid from the outlet is pumped with increased pressure. Care is taken that the outlet volume of the liquid does not back to the inlet.

A single layer vane pump can handle pressures from 17.5 to 70 kg/cm^2 and a two layer pump can handle up to 150 kg/cm^2 of pressure.



Vane Pump

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